

5.3.9 Internal clock timing characteristics

The parameters given in the tables below are derived from tests performed under ambient temperature and supply voltage conditions summarized in Table 18. The curves provided are characterization results, not tested in production.

5.3.9.1 High-speed internal HSI144 oscillator

Table 37. HSI144 oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$f_{\text{HSI}}^{(1)}$	HSI frequency	$V_{\text{DD}} = 3.3 \text{ V}$, $T_J = 30^\circ\text{C}$	144.07	-	145.08	MHz
TRIM ⁽²⁾	USER trimming step	-	-	0.1	0.15	%
USER TRIM COVERAGE ⁽²⁾	USER TRIMMING Coverage positive	80 steps	5.2%	8%	-	
	USER TRIMMING Coverage negative	48 steps	-3.1%	-4.8%	-	
DuCy(HSI) ⁽²⁾	Duty Cycle	-	45		55	
$\Delta_{\text{TEMP}}(\text{HSI})^{(3)}$	HSI oscillator frequency drift over temperature (the reference is 144 MHz.)	$T_J = -20 \text{ to } 130^\circ\text{C}$	-1	-	1	
		$T_J = -40 \text{ to } T_{J\text{max}}^\circ\text{C}$	-1.5	-	1.5	
$\Delta_{V_{\text{DD}}}(\text{HSI})^{(2)(4)}$	HSI oscillator frequency drift with V_{DD} Section 5.3.9.1: High-speed internal HSI144 oscillator ⁽⁵⁾ (the reference is 3.3V)	V_{DD} from 2.7 V to 3.6 V	-	-	- 0.1	μs
$t_{\text{su}}(\text{HSI})^{(3)}$	HSI oscillator start-up time (PSI Off)	-	-	3	4.5	
	HSI oscillator start-up time (PSI On)	-	-	0.5	-	
$t_{\text{stab}}^{(2)}$	stabilization time (PSI OFF from Enable)	+/-1% of target freq	-	-	8	
	stabilization time (PSI ON from Enable)		-	-	0.7	
$I_{\text{DD}}(\text{HSI})^{(2)(6)}$	HSI supply regulation block oscillator power consumption	-	-	91	-	μA
	HSI oscillator power consumption	-	-	28	-	
$N_T \text{ jitter}^{(2)}$	Next transition jitter ⁽⁷⁾ .	On HSIDIV3	-	52	322	ps
$P_T \text{ jitter}^{(2)}$	Paired transition jitter ⁽⁸⁾		-	62	394	
$P_{\text{er}} \text{ jitter}^{(2)}$	Period Jitter standard deviation	On HSIS		15		
		On HSIDIV3		26		

1. Tested in production.
2. Specified by design - Not tested in production.
3. Evaluated by characterization - Not tested in production.
4. $\Delta f_{\text{HSI}} = \Delta_{\text{TEMP}} + \Delta_{V_{\text{DD}}}$
5. These values are obtained by using formula: $(\text{Freq}(3.6 \text{ V}) - \text{Freq}(3.3 \text{ V})) / \text{Freq}(3.3 \text{ V})$ or $(\text{Freq}(3.6 \text{ V}) - \text{Freq}(2.7 \text{ V})) / \text{Freq}(2.7 \text{ V})$.
6. The supply regulation consumption is common to HSI and PSI. (To be counted once if both oscillators are ON).
7. Jitter measurements are performed without clock source activated in parallel. Typical value is standard deviation, Maximum is peak measure on TIE-8 over 36 cycles.
8. Jitter measurements are performed without clock source activated in parallel. Typical value is standard deviation, Maximum is peak measure on TIE-16 over 36 cycles.